

Sean Aidan O'Toole

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EDUCATION

IE University

Master in Business Analytics and Data Science

Madrid, Spain

Sept. 2026 – Aug. 2027 (expected)

University of Texas at San Antonio

B.S. Computer Science, Minor in Mathematics — Honors College

Concentration in Data Science and Software Engineering

San Antonio, TX

Aug. 2022 – May 2026

GPA: 3.90/4.00

SKILLS

Languages: Python, SQL, R, TypeScript, Java

ML & Statistics: scikit-learn, pandas, NumPy, scipy, isotonic regression, clustering, Monte Carlo, backtesting, RAG, LLMs

Data Engineering: ETL design, PostgreSQL, MySQL, SQLite, Parquet, schema normalization, entity resolution, AWS

Tools: Git, Linux, Docker, Jupyter, FastAPI, Matplotlib

EXPERIENCE

Database Developer

The Experts Tribe

May 2025 – Aug. 2025

San Antonio, TX

- Designed and maintained production PostgreSQL, MySQL, and Firebase Firestore systems supporting customer-facing applications.
- Implemented normalized schema structures and data models to support frontend integration and scalability for future feature expansion.
- Optimized SQL queries, indexing strategies, and schema design to reduce latency and improve application responsiveness.

Artificial Intelligence Research Assistant

University of Texas at San Antonio

Oct. 2024 – May 2025

San Antonio, TX

- Built automated feedback loops using Llama-based LLM systems to evaluate and improve software development task output.
- Developed Python data pipelines with BeautifulSoup to collect, clean, and process GitHub repository change requests and review comments at scale.
- Improved model accuracy 60% through iterative experimentation, prompt refinement, and feature engineering on developer-feedback classification.

PROJECTS

World Cup Fantasy — full-stack fantasy platform | *TypeScript, Next.js 15, React 19, PostgreSQL, Drizzle, Vitest*

- Built and shipped a solo full-stack platform in 7 weeks: 53,000 lines of TypeScript across a Next.js/React frontend, a framework-agnostic domain layer, 39 Postgres tables, 22 idempotent migrations, and 51 API routes.
- Designed an immutable-source, derived-everything architecture where standings, head-to-head results, and awards are pure functions over an append-only stat table, so a provider correction propagates everywhere with no cache-invalidation logic.
- Wrote 518 tests across unit, integration (Testcontainers-managed Postgres running real migrations), and component tiers, gated in GitHub Actions CI.

AI Enterprise Data Assistant | *Python, React, FastAPI, RAG, Gemini, Firebase*

- Built a full-stack enterprise application enabling non-technical users to query complex internal databases through natural-language questions.
- Developed a local RAG pipeline grounding Gemini in internal data via multi-stage query rewriting and hybrid retrieval (semantic + BM25) with reranking, cutting index storage and token consumption.
- Architected a decoupled microservices system across separate repositories, with a local cache layer for chat history and Firebase for secure organizational data storage.

AWARDS AND INVOLVEMENT

ACM Media Officer & RowdyHacks Officer | President's List (4x) | Google Data Analytics Certificate